

Poster 2:

Polite, Safe, and Context-Aware Reminder Dialogue for Assistive Home Robots

This research, conducted by Moniruzzaman Akash, focused on developing natural language dialogue systems for assistive home robots that can effectively remind elderly or disabled users of daily tasks while maintaining social appropriateness and safety. The researcher implemented a context-aware reminder dialogue system that understands the user's current activity, emotional state, and preferences to deliver reminders in a polite and non-intrusive manner. The work involved creating algorithms that enable robots to generate contextually relevant, respectful reminders rather than generic automated notifications. The results demonstrated that context-aware dialogue significantly improved user acceptance and compliance with reminders compared to standard notification systems, while also enhancing the perceived politeness and safety of human-robot interactions. This research is significant because it advances assistive robotics by showing that robots can provide more effective care through understanding social context, which is crucial for applications serving vulnerable populations who rely on reminders for medication, appointments, and daily activities.

Poster 3: Assessing LLM Classification Confidence of Reddit Comments Related to Disability

This research, conducted by Nate Thomas, investigated how well Large Language Models (LLMs) can classify and express confidence in their classification of Reddit comments related to disability topics, with particular attention to differences between model architectures from major AI developers. The researchers built a classification system using two leading LLMs, OpenAI's GPT-5.1 and Anthropic's Claude Sonnet-4-20250514, to categorize disability-related Reddit comments and analyzed how each model's confidence scores compared to actual classification accuracy. Key findings revealed significant differences between the two models. GPT-5.1 demonstrated higher confidence scores overall but sometimes overestimated its accuracy, particularly on nuanced disability-related content where understanding cultural context is crucial. In contrast, Claude Sonnet-4-20250514 showed more calibrated confidence metrics, being more conservative in expressing certainty when encountering unfamiliar or sensitive topics. The results demonstrated that confidence calibration varies substantially across different model architectures, with implications for deployment decisions in sensitive domains. This work is significant because it highlights how different AI companies' design choices and training methodologies produce models with different reliability profiles, a critical consideration when selecting which LLM to use for applications affecting vulnerable communities like those with disabilities. The research contributes to the growing field of AI trustworthiness by showing that not all high-performing models are equally reliable in their confidence assessments, and that architectural differences between providers matter significantly when accuracy and appropriate uncertainty expression are paramount.